



3.1.5 Wrap-Up: Growth Mindset

Complete each statement based on today's lesson.

1. Today I am still unsure about ...

2. I will try to build my confidence by ...

Unit 3, Lesson 2: Writing Equations of Lines



Benchmarks

MA.912.AR.2.2 Write a linear two-variable equation to represent relationships between quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.



Learning Targets

- ☐ I can determine the slope of a line from its graph.
- ☐ I can write the equation of a line given its graph.
- ☐ I can rearrange the equations of a line from one form into another form.



3.2.1 Warm-Up: Growth Mindset

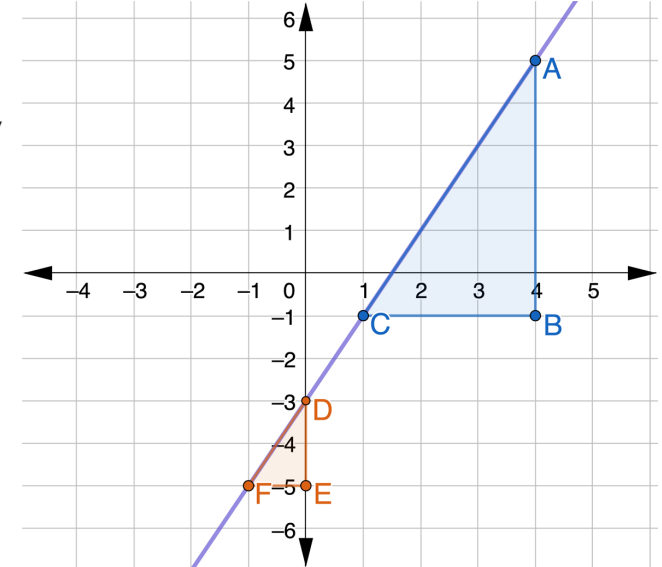
Dr. Jo Boaler is a professor in mathematics education. She has been quoted saying, "You may have been told or led to believe that some people are 'math people' and some aren't ... or that math isn't for you. Well, it turns out, that's not true. Everyone is born with the innate ability to do well in math. And whether you do well or not comes from the experience you've had and the beliefs you hold."

1. What has your experience been like when learning math?
2. Do you think that you are a math person? Explain.



3.2.2 Exploration: Determining the Slope of a Line From Its Graph

Similar triangles can be used to determine the slope of a line by comparing the ratio of vertical lengths to horizontal lengths of the triangles.



1. For each triangle, label the ordered pairs of each vertex.
2. Use the similar triangles shown on the graph to complete the following with your partner.
 - a. Complete the table.

Triangle	Vertical Leg Length	Horizontal Leg Length	Ratio
			$\frac{\text{vertical leg length}}{\text{horizontal leg length}}$
<i>ABC</i>			
<i>DEF</i>			

b. Complete the statement.

The ratio $\frac{\text{vertical leg length}}{\text{horizontal leg length}}$ for the blue triangle is

- ☐ greater than
- ☐ equivalent to
- ☐ less than

the ratio $\frac{\text{vertical leg length}}{\text{horizontal leg length}}$ for the

orange triangle.

The slope of a line is the ratio of the change in the vertical direction to the change in the horizontal direction, often expressed as the $\frac{\text{change in } y}{\text{change in } x}$ or $\frac{\Delta y}{\Delta x}$.

3. Complete the statement.

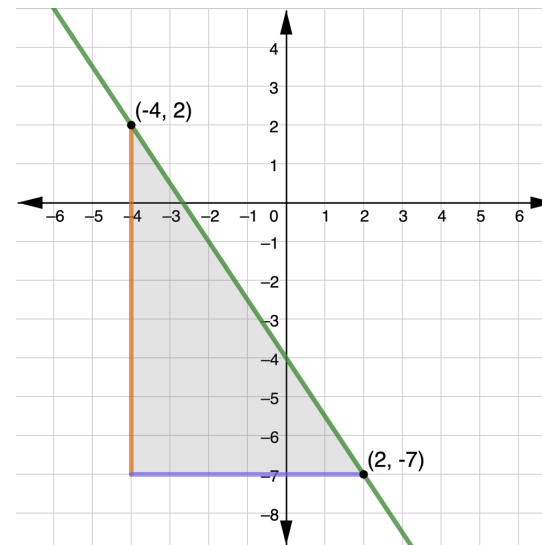
In the equation $\text{slope} = \frac{\text{change in } y}{\text{change in } x}$, the ☐ change in y
☐ change in x

relates to finding the length of the vertical leg of the

triangle and the ☐ change in y
☐ change in x relates to finding the

length of the horizontal leg of the triangle.

4. Mei and Kwame both drew the same triangle using the points $(-4, 2)$ and $(2, -7)$ on the line to determine the slope, but they performed the calculations differently. Their partial work is shown.



a. With your partner, complete each student's work.

Mei's Work	Kwame's Work
$\text{slope} = \frac{\text{change in } y}{\text{change in } x}$	$\text{slope} = \frac{\text{change in } y}{\text{change in } x}$
$\text{slope} = \frac{-7 - (2)}{2 - (-4)}$	$\text{slope} = \frac{2 - (-7)}{-4 - (2)}$
$\text{slope} =$	$\text{slope} =$

b. What is the slope of the line, in simplest form?

- c. Mei compared her work to Kwame's and said, "The order in which we subtracted the values doesn't matter. We found the same slope." Explain why you agree or disagree.

- d. Kwame said, "The order in which we subtracted doesn't matter, but whichever coordinate pair you start with to find the change in y , you have to also start with its x -value when finding the change in x ." What do you think Kwame meant?

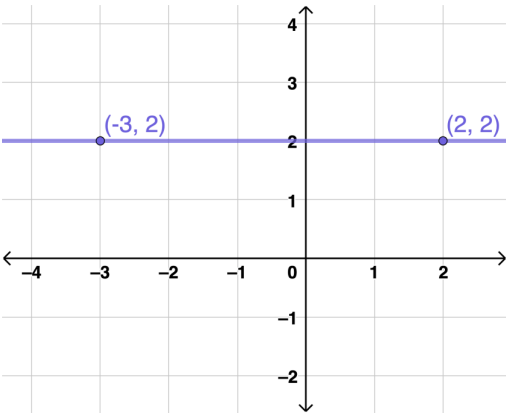
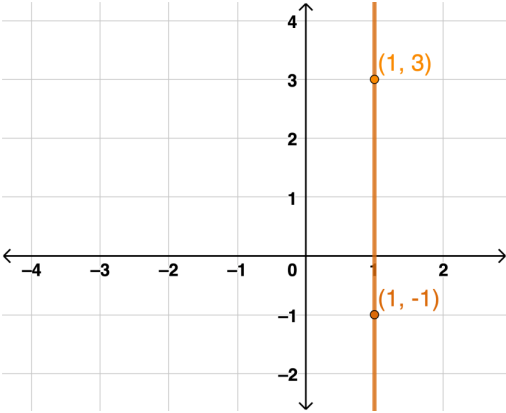


3.2.3 Bring It Together: Determining the Slope of a Line From Its Graph

The slope of any line can be determined from a graph using any two points on the line and calculating the $\frac{\text{change in } y}{\text{change in } x}$.

Four graphs and the calculations used to determine their slope, m , are shown.

Graph	Calculation	Slope
	$m = \frac{4 - 1}{2 - 0} = \frac{3}{2}$	$\frac{3}{2}$
	$m = \frac{1 - (-2)}{-4 - 0} = \frac{3}{-4}$	$-\frac{3}{4}$

	$m = \frac{2 - 2}{2 - (-3)} = \frac{0}{5}$	0
	$m = \frac{-1 - 3}{1 - 1} = \frac{-4}{0}$	undefined

1. Explain why the slope of the vertical line in the table is undefined.

2. Use the graphs and their slopes to complete the summary.

The slope of a line can be positive, negative, 0, or undefined. The slope is positive when the line

☐ is increasing, or rising,
☐ is decreasing, or falling,

from left to right. The slope is

negative when the line

☐ is increasing, or rising,
☐ is decreasing, or falling,

from

left to right. The slope of

☐ any vertical line
☐ any horizontal line

is 0

because

☐ the change in x is 0.

☐ it is neither increasing nor decreasing from left to right.

The slope of

☐ any vertical line
☐ any horizontal line

is undefined because

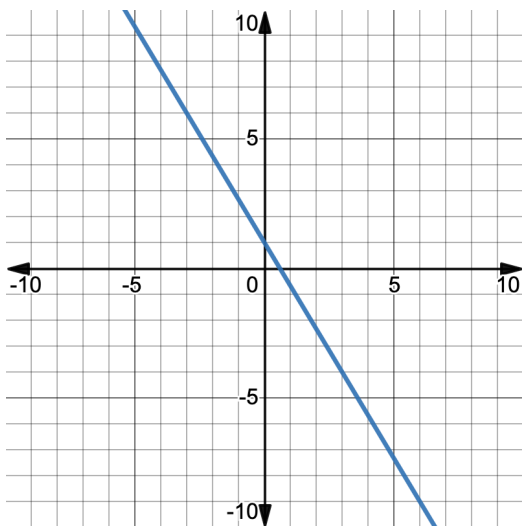
☐ the change in x is 0.

☐ it is neither increasing nor decreasing from left to right.



3.2.4 Guided Instruction: Writing the Equation of a Line From a Graph

1. The graph of a line is shown.



- Discuss with your partner which values you would need to find in order to write the equation of the line in point-slope form.
- Determine the slope of the line.
- Jack and Izzy planned to use different points on the line to write the equation in point-slope form. Use the points each student identified to write two different equations for the line in point-slope form.

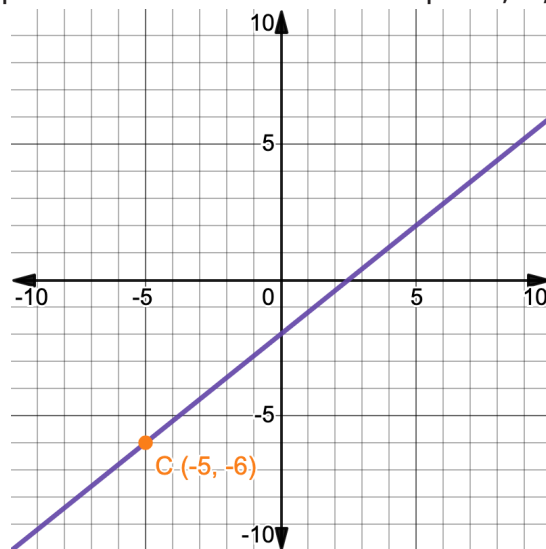
Jack's Equation using the point $(-3, 6)$	Izzy's Equation using the point $(3, -4)$



- d. Rewrite Jack's equation in slope-intercept form.

- e. Rewrite Izzy's equation in standard form.

2. The graph of a line is shown with a point, C , labeled.



Write the equation of the line in point-slope form using the point labeled on the graph.

3. Fatima and Isaiah used different strategies to rewrite the equation from Problem 2 in standard form. Their first steps are shown in blue.

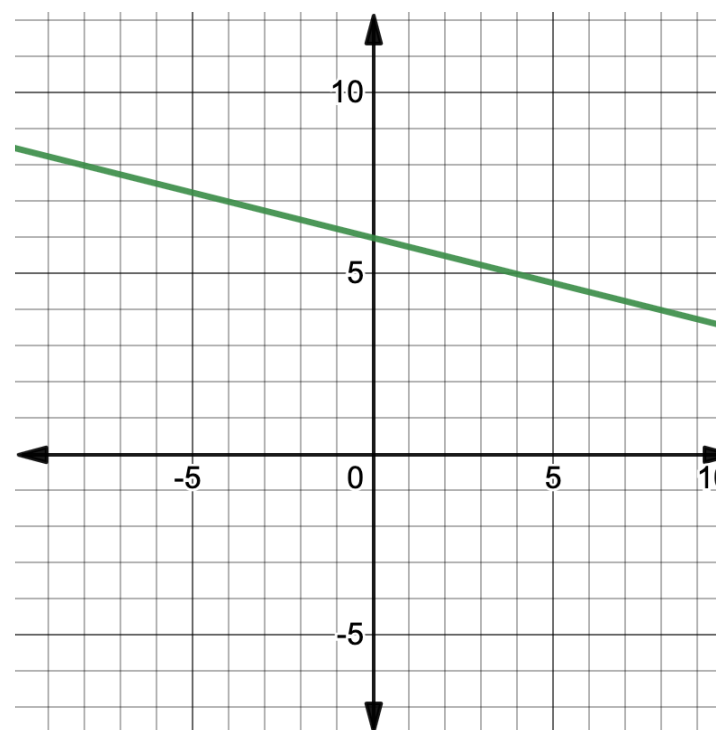
Fatima's Work	Isaiah's Work
$(y + 6) = \frac{4}{5}(x + 5)$ $5[(y + 6) = \frac{4}{5}(x + 5)]$ $5y + \underline{\hspace{2cm}} = 4(x + 5)$	$(y + 6) = \frac{4}{5}(x + 5)$ $(y + 6) = \frac{4}{5}x + 4$

- Discuss with your partner what each student did in their first step. Summarize your discussion.
- Complete each student's work to rewrite the equation in standard form in the space provided above.



3.2.5 Try It: Writing the Equation of a Line From a Graph

1. The graph of a line is shown.



- Write the equation of the line in point-slope form.
- Rewrite the equation from Part A in standard form.



3.2.6 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate how you feel about each learning target.

- a. I can determine the slope of a line from its graph.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
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- b. I can write the equation of a line given its graph.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
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- c. I can rearrange equations of a line from one form into another form.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
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Unit 3, Lesson 3: Writing Equations of Lines From Tables or Graphs



Benchmarks

MA.912.AR.2.2 Write a linear two-variable equation to represent relationships between quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.



Learning Targets

- ☐ I can write the equation of a line, given a graph or a table.
- ☐ I can write the equation of a line, given a graph or a table in a real-world context.
- ☐ I can rearrange the equations of a line from one form into another form.